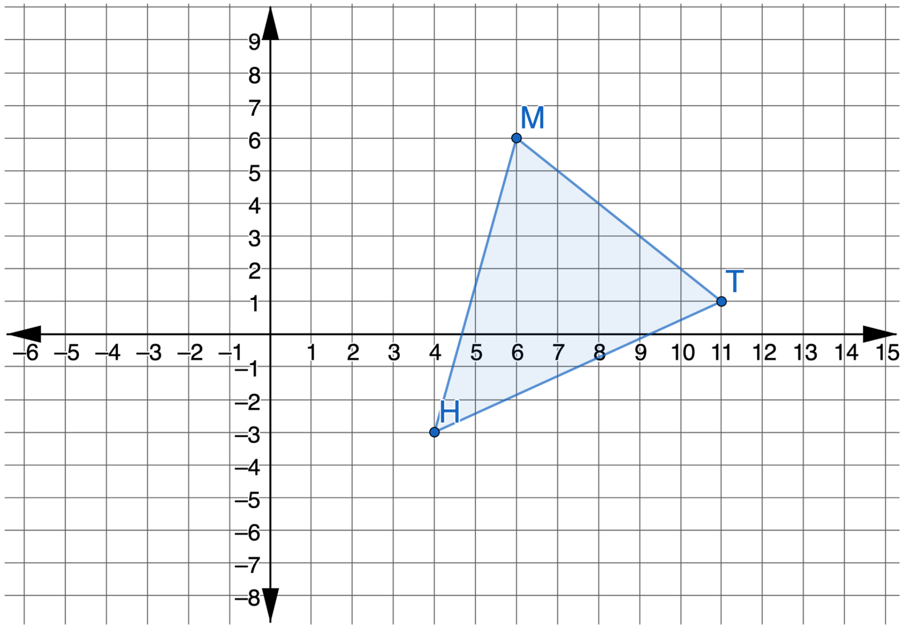
4.4.5 Wrap-Up: Ticket Out the Door

1. Triangle HMT is shown on the coordinate grid.



Draw ΔBWF, which is the result of translating ΔHMT left 3 and up $3,$ then rotating the result $90°$ clockwise about the origin.

Unit 4, Lesson 5: Algebraic Descriptions of a Sequence of Transformations

Benchmarks

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

MA.912.GR.2.5 Given a geometric figure and a sequence of transformations, draw the transformed figure on a coordinate plane.

Learning Targets

- ☐ I can draw an image given a pre-image and an algebraic description of a sequence of transformations on a coordinate plane.
- ☐ I can write a transformation on the coordinate plane using coordinate notation with algebraic descriptors.

4.5.1 Warm-Up: Algebraic Descriptions of Transformations

1. Use your notes to complete the algebraic descriptions in the table.

The Effect of Transformations on Coordinates

a.

Rotations Clockwise About the Origin		
Rotations of $90°$	**Rotations of $180°$**	**Rotations of $270°$**
Change the x-coordinate to its opposite, then switch the coordinates	Change both coordinates to their opposite	Change the y-coordinate to its opposite, then switch the coordinates
$(x, y) \rightarrow (\_\_\_\_, \_\_\_\_)$	$(x, y) \rightarrow (\_\_\_\_, \_\_\_\_)$	$(x, y) \rightarrow (\_\_\_\_, \_\_\_\_)$

Rotations Counterclockwise About the Origin		
Rotations of $90°$	**Rotations of $180°$**	**Rotations of $270°$**
Change the y-coordinate to its opposite, then switch the coordinates	Change both coordinates to their opposite	Change the x-coordinate to its opposite, then switch the coordinates
$(x, y) \rightarrow (\_\_\_\_, \_\_\_\_)$	$(x, y) \rightarrow (\_\_\_\_, \_\_\_\_)$	$(x, y) \rightarrow (\_\_\_\_, \_\_\_\_)$

b.

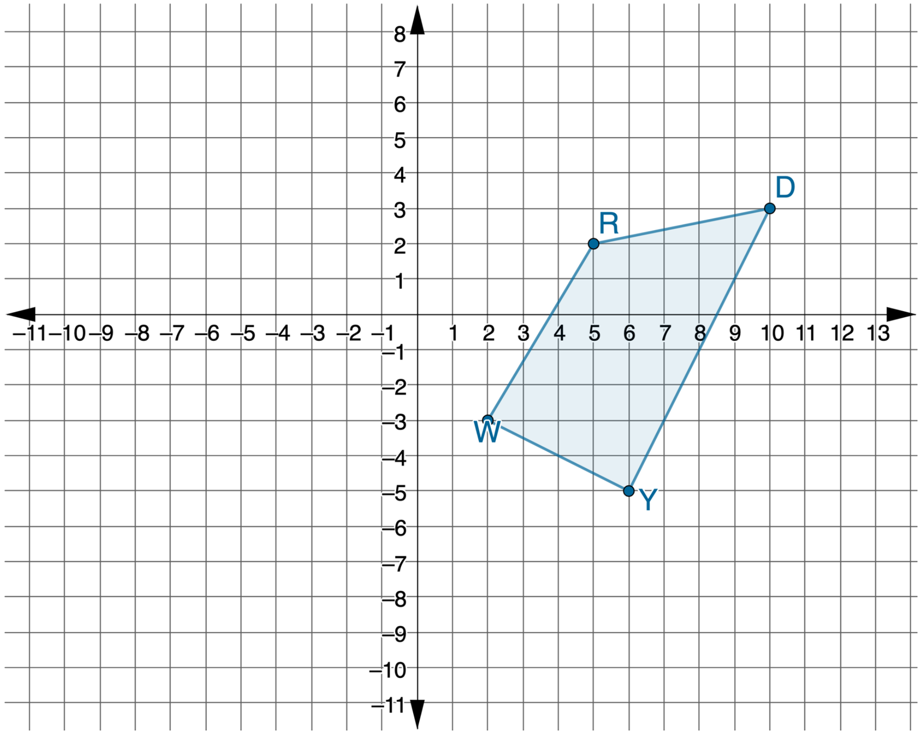
Translations	
Right	**Left**
Add to the x-coordinate	Subtract from the x-coordinate
$(x, y) \rightarrow (\_\_\_\_, \_\_\_\_)$	$(x, y) \rightarrow (\_\_\_\_, \_\_\_\_)$
Up	**Down**
Add to the y-coordinate	Subtract from the y-coordinate
$(x, y) \rightarrow (\_\_\_\_, \_\_\_\_)$	$(x, y) \rightarrow (\_\_\_\_, \_\_\_\_)$

c.

Reflections	
x-axis	**y-axis**
The y-coordinate changes to its opposite.	The x-coordinate changes to its opposite.
$(x, y) \rightarrow (\_\_\_\_, \_\_\_\_)$	$(x, y) \rightarrow (\_\_\_\_, \_\_\_\_)$
$y = x$	**$y = -x$**
The coordinates switch.	The coordinates switch and become the opposite.
$(x, y) \rightarrow (\_\_\_\_, \_\_\_\_)$	$(x, y) \rightarrow (\_\_\_\_, \_\_\_\_)$

4.5.2 Guided Instruction: Applying Algebraic Descriptions

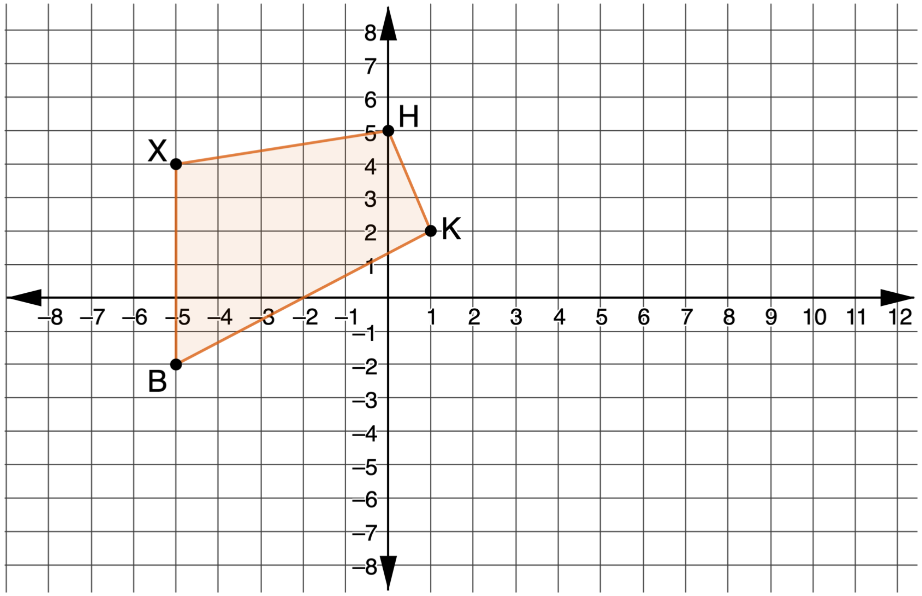
1. Quadrilateral $RDYW$ is shown on the coordinate grid.



Apply the sequence of transformations to quadrilateral $RDYW$. Draw the result of each transformation on the graph. Assume all rotations are centered at the origin.

- $(x, y) \to (-x, y)$ Label as $R'D'Y'W'$.
- $(x, y) \to (-y, x)$ Label as $R''D''Y''W''$.

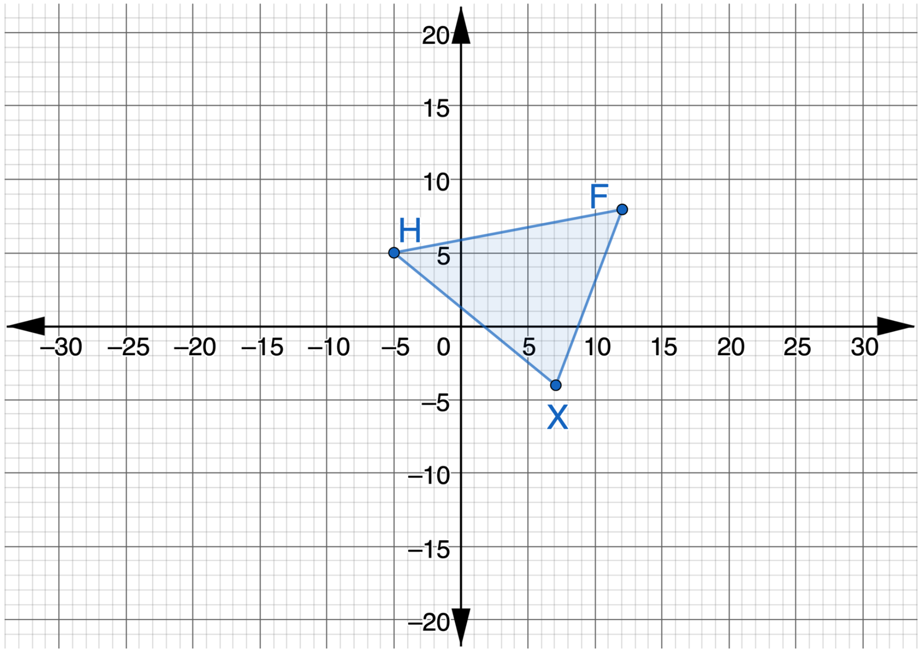
2. Quadrilateral $BXHK$ is shown on the coordinate grid.



Apply the series of transformations to quadrilateral $BXHK$. Draw the result of each transformation on the graph. Assume all rotations are centered at the origin.

- $(x, y) \to (x + 10, y - 4)$ Label as $B'X'H'K'$.
- $(x, y) \to (x, -y)$ Label as $B''X''H''K''$.

3. Triangle HFX is shown on the coordinate grid.

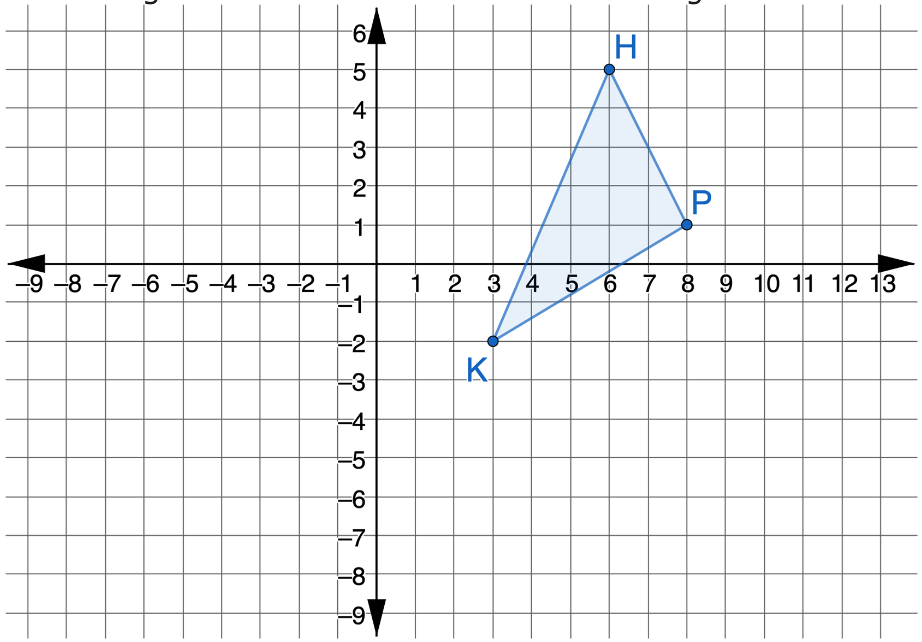


Perform the sequence of transformations on ΔHFX. Assume all rotations are centered at the origin. Label the result of all three transformations as ΔTMR.

- $(x, y) \to (x - 22, y + 9)$
- $(x, y) \to (-x, y)$
- $(x, y) \to (-x, -y)$

4.5.3 Guided Practice: Applying Algebraic Descriptions

1. Triangle PKH is shown on the coordinate grid.

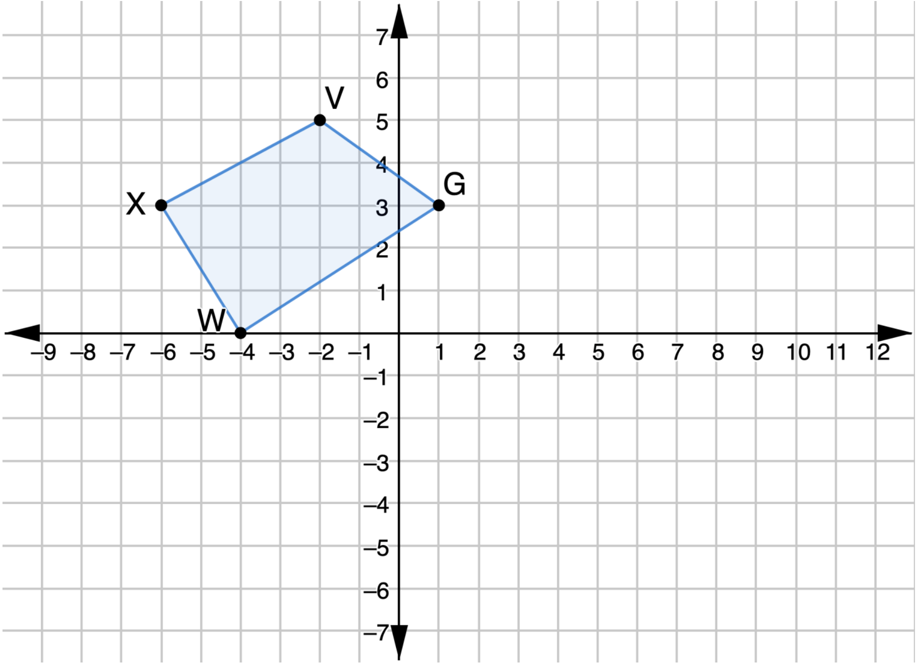


a. Perform the sequence of transformations on ΔPKH. Assume all rotations are centered at the origin. Label the result of transformations as ΔYTM.

- $(x, y) \to (x, -y)$
- $(x, y) \to (x - 9, y - 3)$

b. Check your answer with your partner.

2. Quadrilateral $XVGW$ is shown on the coordinate grid.

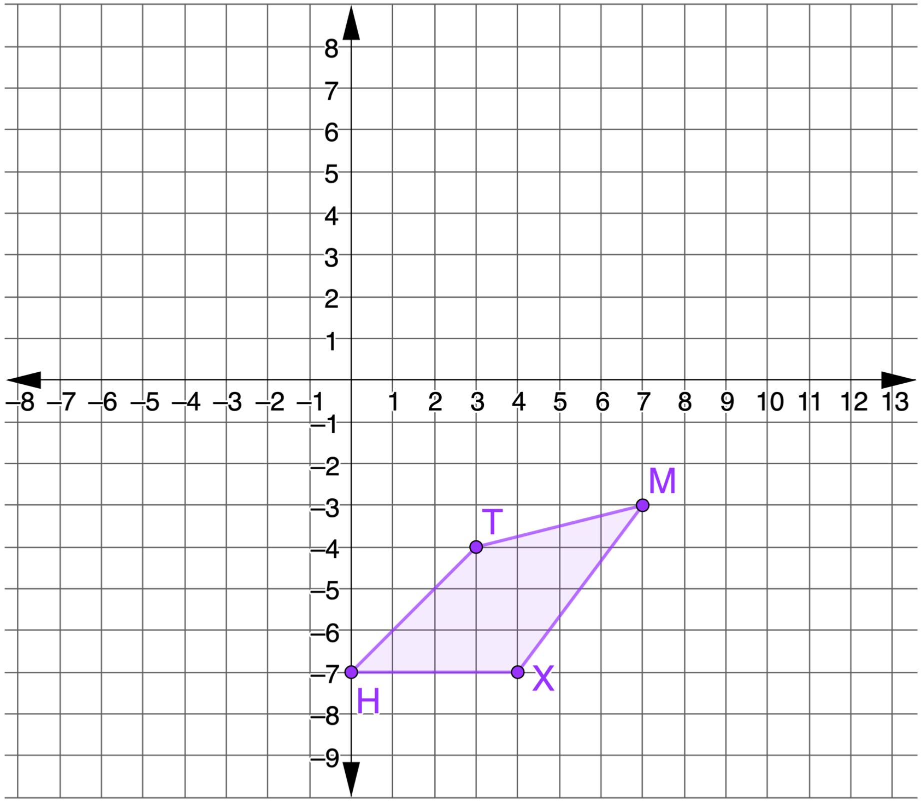


Perform the series of transformations on quadrilateral $XVGW$. Assume all rotations are centered at the origin. Label the result of transformations as quadrilateral $BHMK$.

- $(x, y) \rightarrow (-x, -y)$
- $(x, y) \rightarrow (y, -x)$
- $(x, y) \rightarrow (x + 4, y - 1)$

4.5.4 Your Turn!

1. Quadrilateral $XHTM$ is shown on the coordinate grid.

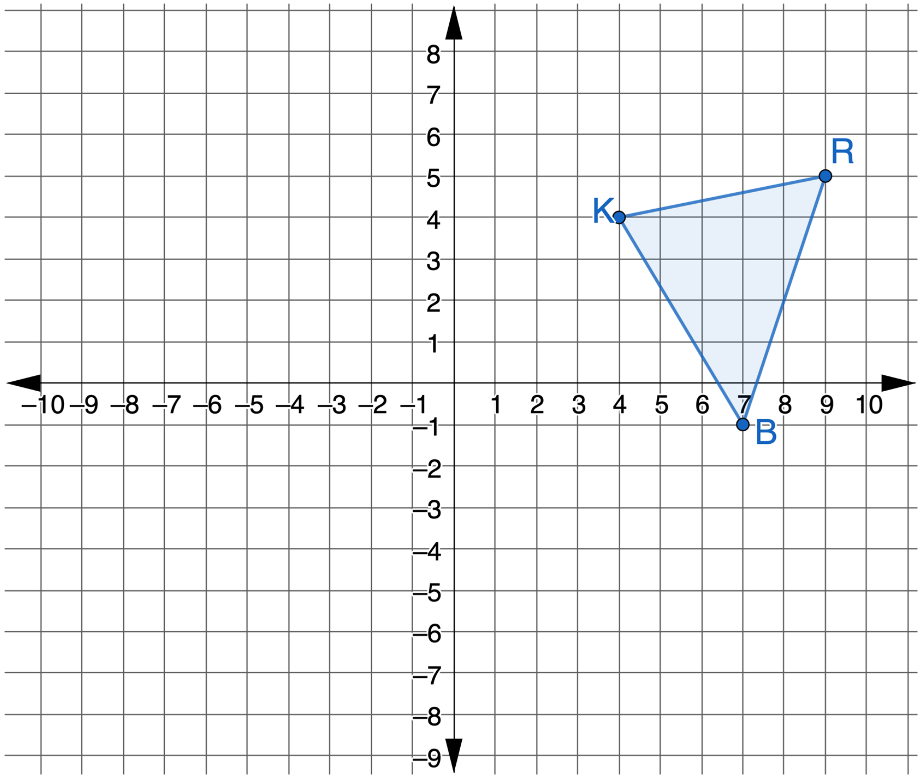


Perform the sequence of transformations on quadrilateral $XHTM$. Assume all rotations are centered at the origin. Label the result of the transformations as quadrilateral $WQPZ$.

- $(x, y) \rightarrow (-x, y)$
- $(x, y) \rightarrow (-y, x)$
- $(x, y) \rightarrow (x + 4, y - 1)$

4.5.5 Wrap-Up: Ticket Out the Door

1. Triangle KRB is shown on the coordinate grid.



Perform the sequence of transformations on ΔKRB. Assume all rotations are centered at the origin. Label the result of the transformations as ΔTLS.

- $(x, y) \rightarrow (x - 12, y + 3)$
- $(x, y) \rightarrow (x, -y)$

Unit 4, Lesson 6: Isometry

Benchmarks

MA.912.GR.2.3 Specify a sequence of transformations that will map a given figure onto itself, or onto another congruent or similar figure.

MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.

Learning Targets

- ☐ I can describe a sequence of transformations that map one figure onto another congruent figure.
- ☐ I can draw an image given a preimage and a sequence of transformations on a coordinate plane.